

```
In [2]: i = 1
        while i<=4:
            j = 0
            while j <= 3:
                print(i * j,end=" ")
                j += 1
            print()
            i += 1
```

```
0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12
```

```
In [3]: # for loop
        name = [1,2,3,'hello']

        for i in name:
            print(i)
```

```
1
2
3
hello
```

```
In [5]: range(5)
```

```
Out[5]: range(0, 5)
```

```
In [4]: for i in range(5):
        print(i)
```

```
0
1
2
3
4
```

```
In [6]: for i in range(1,10,3):
        print(i)
```

```
1
4
7
```

```
In [9]: for i in range(1,10,5):
        print(i)
```

```
1
6
```

```
In [ ]: # multiplication table of 5
        for i in range(1,51):
            if i % 5 == 0:
                print(i)
```

5
10
15
20
25
30
35
40
45
50

```
In [ ]: # multiplication table of 3  
for i in range(1,31):  
    if i % 3 == 0:  
        print(i)
```

3
6
9
12
15
18
21
24
27
30

```
In [12]: # number which is not divisible by 3  
for i in range(1,31):  
    if i % 3 != 0:  
        print(i)
```

1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29

```
In [13]: # multiplication table of 5 number which is not divisible by 5  
for i in range(1,51):  
    if i % 5 != 0:  
        print(i)
```

1
2
3
4
6
7
8
9
11
12
13
14
16
17
18
19
21
22
23
24
26
27
28
29
31
32
33
34
36
37
38
39
41
42
43
44
46
47
48
49

```
In [14]: for i in range(1,11):  
         print(i)
```

1
2
3
4
5
6
7
8
9
10

```
In [15]: for i in range(1,11):  
         if i == 5:  
             break  
         print(i)
```

1
2
3
4

```
In [16]: for i in range(1,11):  
         if i == 5:  
             continue  
         print(i)
```

1
2
3
4
6
7
8
9
10

```
In [17]: for i in range(1,10):  
         pass
```

```
In [18]: for i in range(1,11):  
         if i == 5:  
             continue  
         print('hello',i)
```

hello 1
hello 2
hello 3
hello 4
hello 6
hello 7
hello 8
hello 9
hello 10

```
In [21]: for i in range(1,51):  
         if i % 3 == 0 or i % 5 == 0:  
             continue  
         print(i)  
         print('end')
```

```
1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
end
```

Patterns

```
In [22]: for i in range(5):
          print('#')
```

```
#
#
#
#
#
```

```
In [24]: for i in range(1,5):
          print('####')
```

```
####
####
####
####
```

```
In [1]: for i in range(1,51):

          if (i%2 == 0):
              #print('even')
              continue
          else:
```

```
print(i)
print('bye')
```

```
1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
31
33
35
37
39
41
43
45
47
49
bye
```

```
In [2]: for j in range(4):
        print('#', end = " ")
```

```
# # # #
```

```
In [3]: for j in range(4):
        print('#', end=" ")

        print()

        for j in range(4):
            print('#', end=" ")
```

```
# # # #
# # # #
```

```
In [4]: for i in range(4):
        for j in range(4):
            print('#', end=" ")
        print()
        # please use debug mode in pycharm
```

```
# # # #
# # # #
# # # #
# # # #
```

In []: